



4th Champéry Power Conference, 9 February 2026, Switzerland

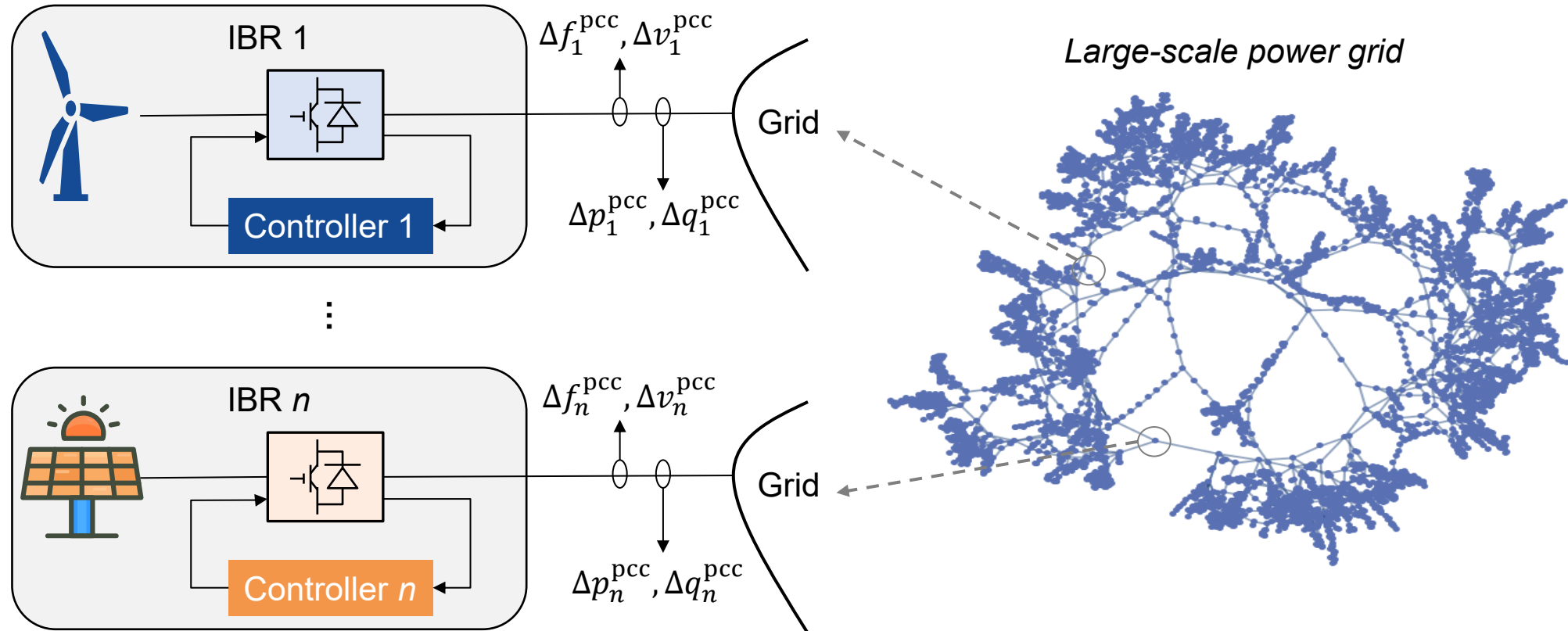
Quantifying Stability Conditions for Grid-Forming and Grid-Following Heterogeneous Power Systems

Xiuqiang He, Xiang Zhu, and Hua Geng
Tsinghua University



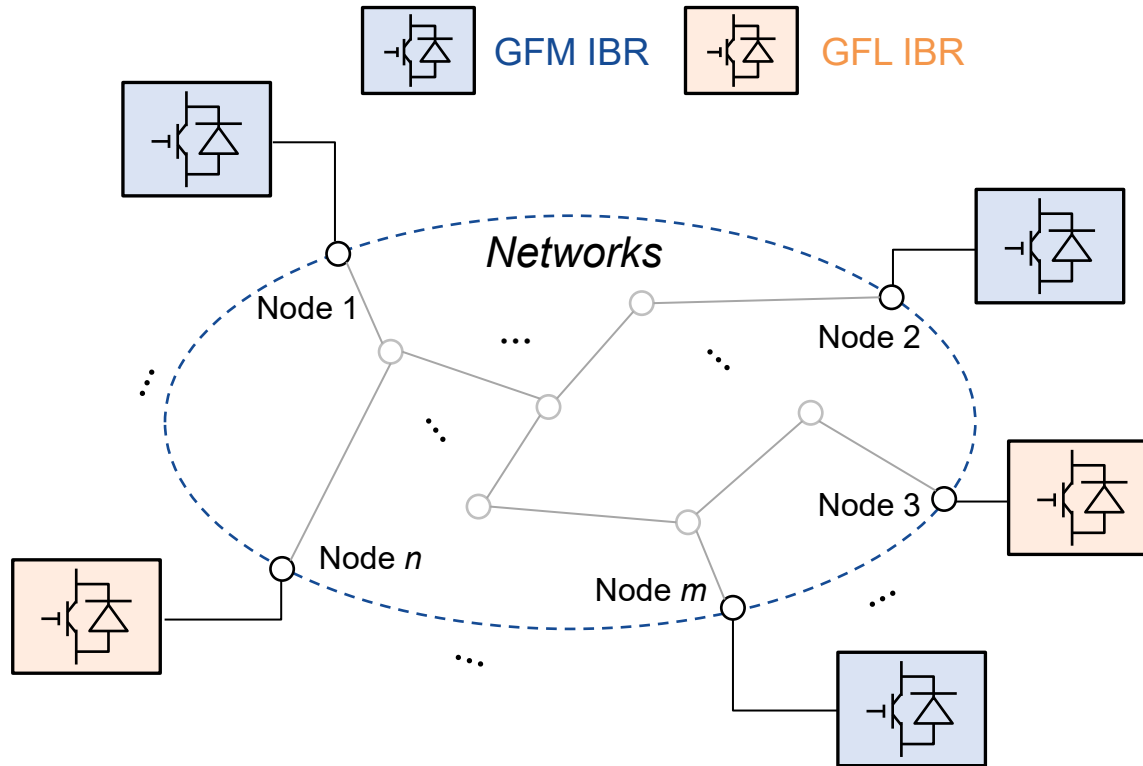
清华大学
Tsinghua University

Quantifying Stability Conditions for IBRs



- Inverter-based resources (IBRs)-dominated power systems suffer from many stability issues
- **Quantifying stability conditions** can provide guidelines for developing IBRs' dynamic response specifications (grid codes) and control design

Heterogeneous Power Systems



- GFM controls:

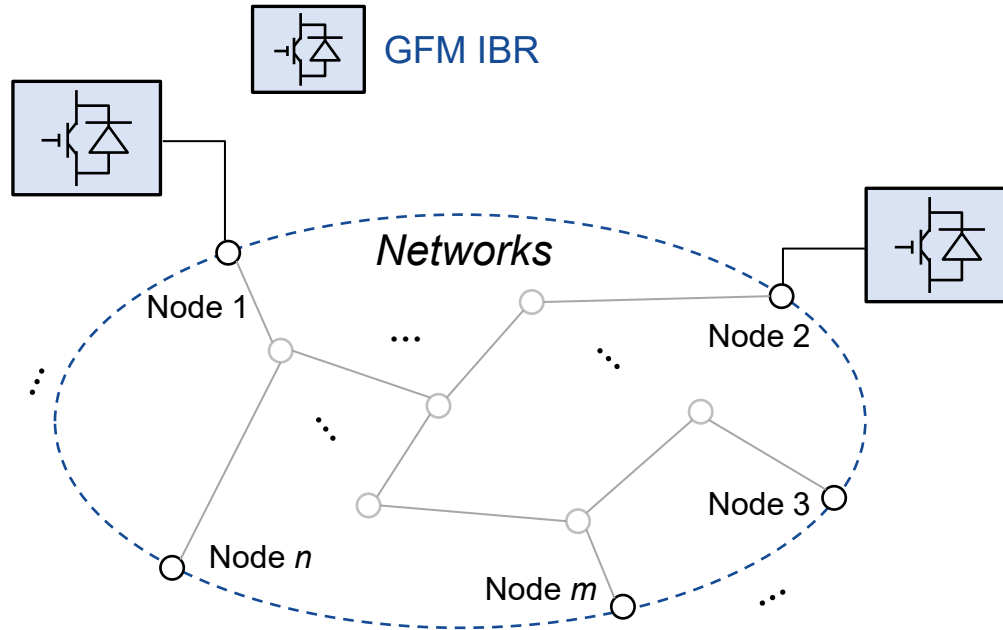
$$\begin{cases} D_i^{g,p}(s) = \frac{\Delta\theta_i^g(s)}{\Delta P_i^g(s)} \\ D_i^{g,q}(s) = \frac{\Delta v_i^g(s)}{\Delta Q_i^g(s)} \end{cases}$$

- GFL controls

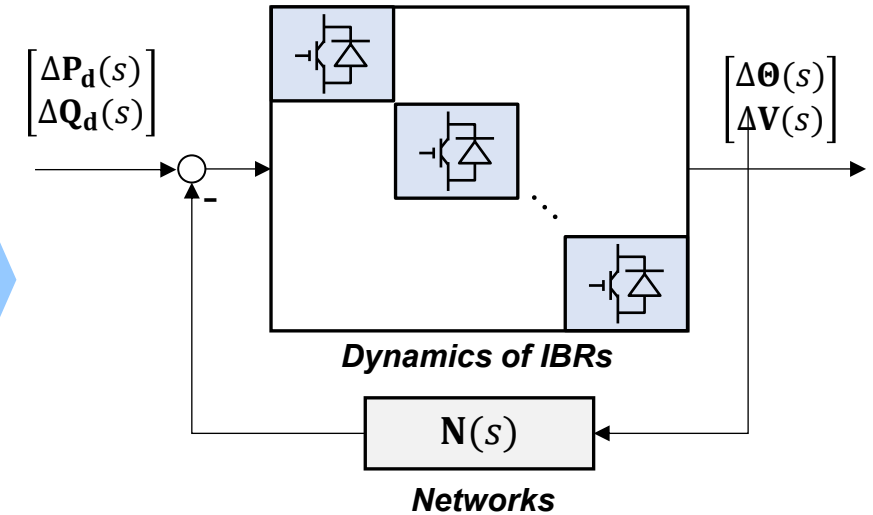
$$\begin{cases} D_i^{f,p}(s) = \frac{\Delta P_i^f(s)}{\Delta\theta_i^f(s)} \\ D_i^{f,q}(s) = \frac{\Delta Q_i^f(s)}{\Delta v_i^f(s)} \end{cases}$$

- Grid-forming (**GFM**) control and grid-following (**GFL**) control are two typical control modes
- Future IBR-dominated power systems will be **GFM-GFL-mixed**

Centralized Analysis Framework

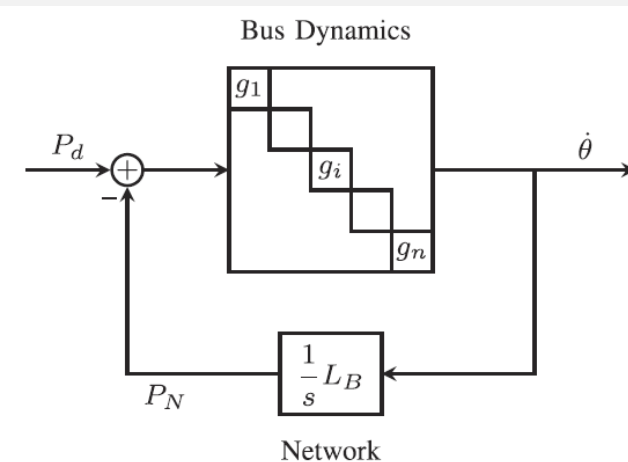


GFM-Only Homogeneous Power System

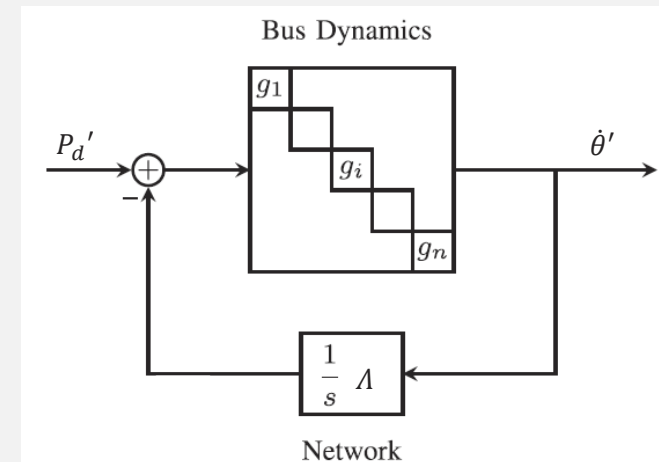


- A decoupling analysis approach based on eigen-decomposition [1]

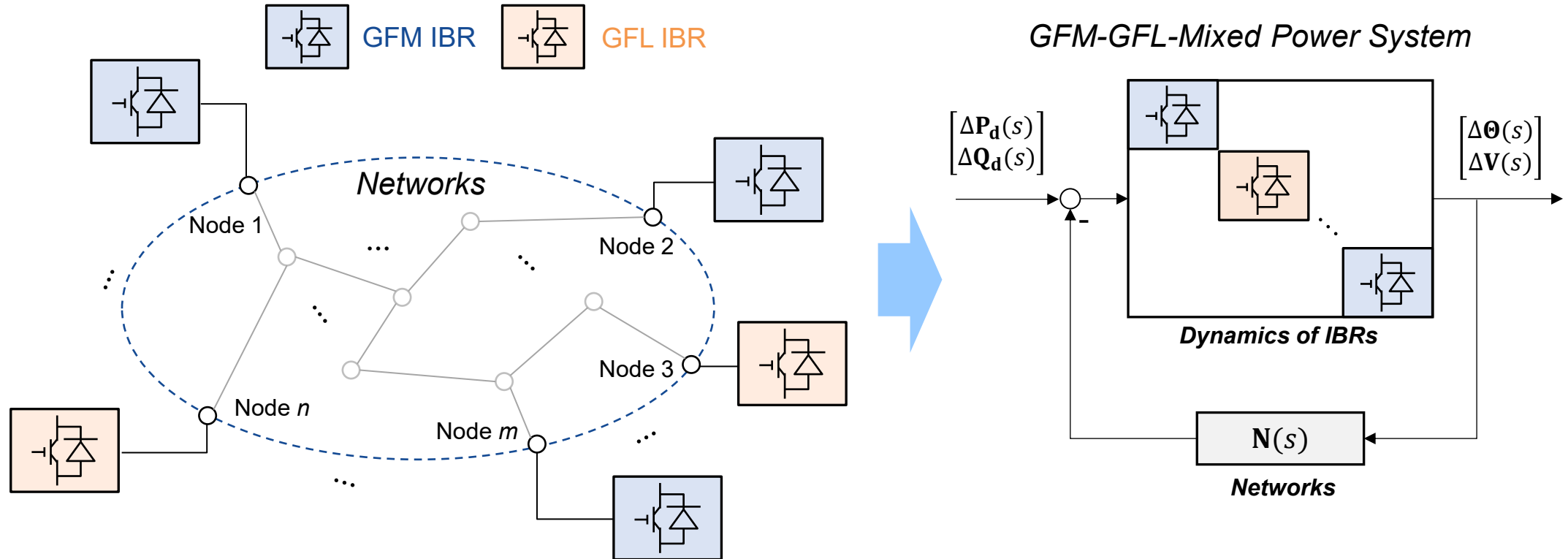
[1] Jiang, Y., Pates, R., & Mallada, E. (2020). Dynamic droop control in low-inertia power systems. *IEEE TAC*, 66(8), 3518-3533.



$$W^{-1}L_B W = \Lambda$$



Decentralized Analysis Framework



Why Decentralization? **Massiveness, Heterogeneity, Model-agnosticism**

- Passivity (with multipliers or loop shifting)
- Small gains (quite conservative)

Only stability (w/o ensuring damping ratio)

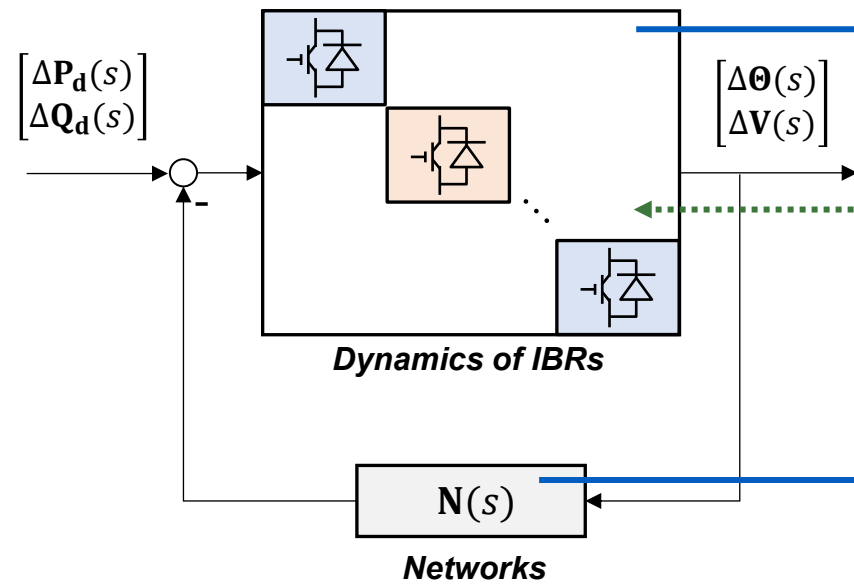
Today → A new decentralized approach

- Stability
- Damping ratio (pole distribution)

Outline



Mixed GFM-GFL Heterogeneous Power System



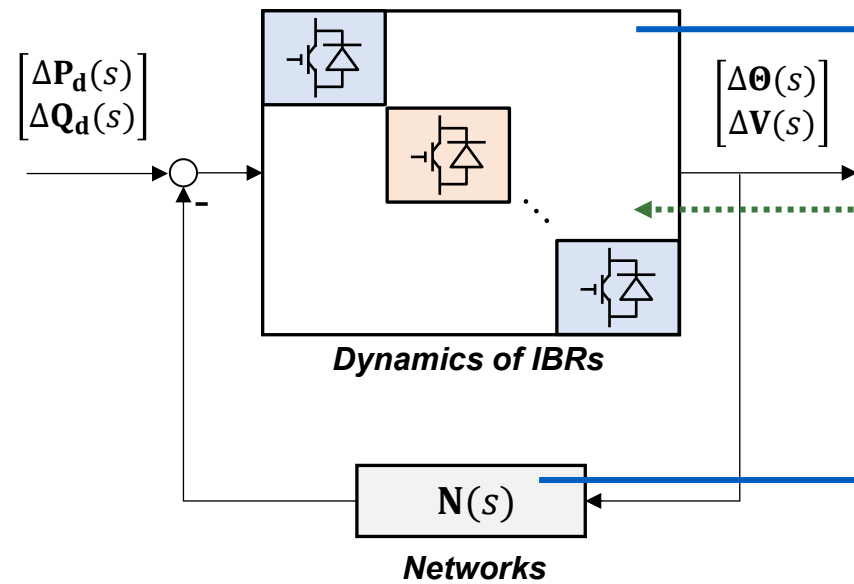
Part I: Theoretical Analysis

Part II: Applications

Outline



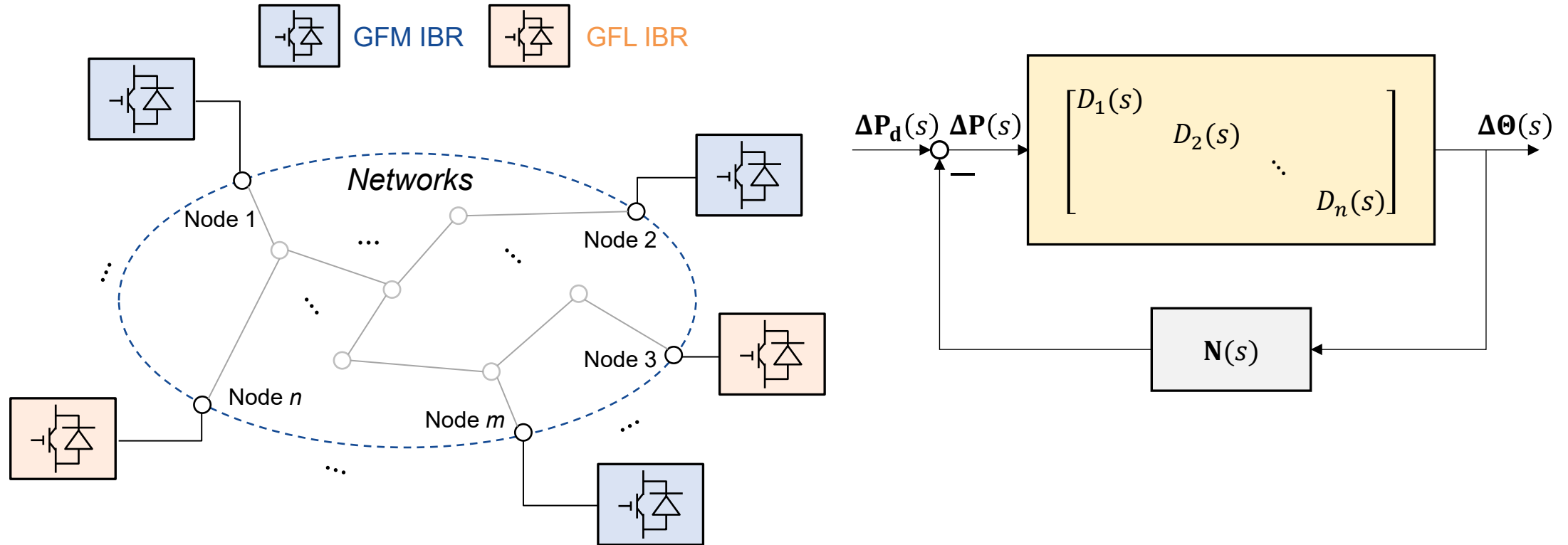
Mixed GFM-GFL Heterogeneous Power System



Part I: Theoretical Analysis

Part II: Applications

Closed-Loop Modeling for Heterogeneous System



Modeling assumptions:

- Considering small-signal frequency dynamics + constant voltages
- Kron-reduced network, retaining GFM/GFL nodes only

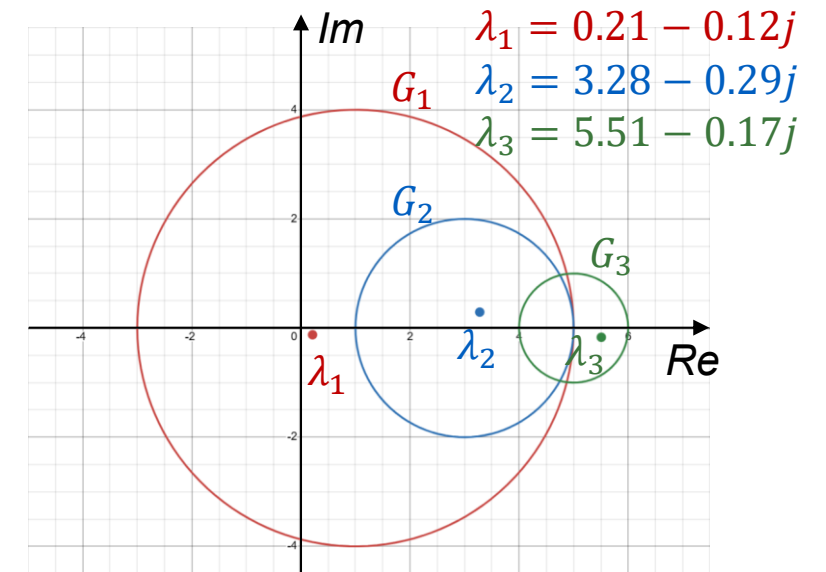


Allowing for more general extensions

Preliminaries

How to estimate the distribution of eigenvalues of a matrix?

$$\mathbf{M} = \begin{bmatrix} 1 & -2 & -2 \\ -1 & 3 & j \\ 0 & -j & 5 \end{bmatrix} \xrightarrow{\text{Find circles}} \begin{cases} G_1 = |\lambda_1 - 1| \leq 4 \\ G_2 = |\lambda_2 - 3| \leq 2 \\ G_3 = |\lambda_3 - 5| \leq 1 \end{cases} \xrightarrow{\text{Estimate the eigenvalues}}$$



Gershgorin Theorem: For a matrix $\mathbf{M}$, its eigenvalues must lie within the region formed by all the “Gershgorin Circles”.

$$G_i = \left\{ \lambda_i \in \mathbb{C} \mid \underbrace{|\lambda_i - M_{ii}|}_{\text{diagonal elements}} \leq \underbrace{\sum_{j \neq i} |M_{ij}|}_{\text{sum of moduli of off-diagonal elements}} \right\}$$

diagonal
elements

sum of moduli of
off-diagonal
elements

Preliminaries

Gershgorin Theorem: For a matrix $\mathbf{M}$, its eigenvalues must lie within the region formed by all the “Gershgorin Circles”.

$$|M_{ii}| > \sum_{j \neq i} |M_{ij}|, i = 1, 2, \dots, n$$

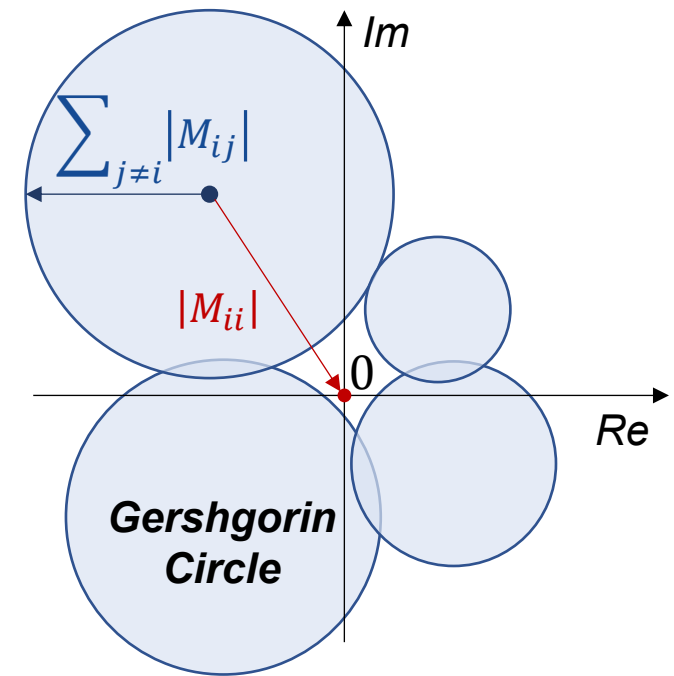
$$\Rightarrow \lambda_i \neq 0, i = 1, 2, \dots, n$$

Sufficient
Condition

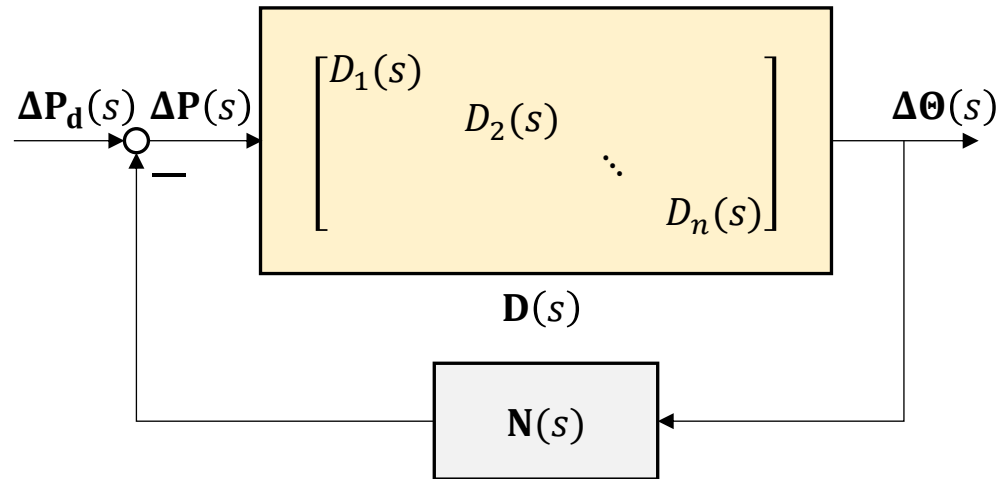
Non-singular: $\det(\mathbf{M}) \neq 0$

Levy-Desplanques Theorem: If $\mathbf{M}$ satisfies the strictly diagonally dominant matrix (DDM) condition, then $\mathbf{M}$ is non-singular.

$$G_i = \left\{ \lambda_i \in \mathbb{C} \mid |\lambda_i - M_{ii}| \leq \sum_{j \neq i} |M_{ij}| \right\}$$



Proposed Local Gain Condition (LGC)



- Characteristic equation: $\det(\mathbf{I} + \mathbf{D}(s)\mathbf{N}(s)) = 0$
 $\Rightarrow \det(\mathbf{D}(s)) \det((\mathbf{D}(s))^{-1} + \mathbf{N}(s)) = 0$
- DDM (diagonally dominant matrix) conditions:
 $\Rightarrow \underbrace{|(D_i(s))^{-1} + N_{ii}(s)|}_{\text{Decoupled local gain}} > \underbrace{\sum_{j \neq i} |N_{ij}(s)|}_{\text{Coupling gain}} > 0, \forall s \in \Gamma$
 $\Rightarrow \det(\mathbf{I} + \mathbf{D}(s)\mathbf{N}(s)) \neq 0, \forall s \in \Gamma$ (e. g., RHP)

Local Gain Condition (LGC): Assume that node dynamics $|\mathbf{D}(s)|$ are non-singular within a prohibited domain Γ (e.g., RHP). Thus, **if the LGC condition is satisfied in Γ for each IBR, no closed-loop poles lie within this domain.**

From Domain to Boundary

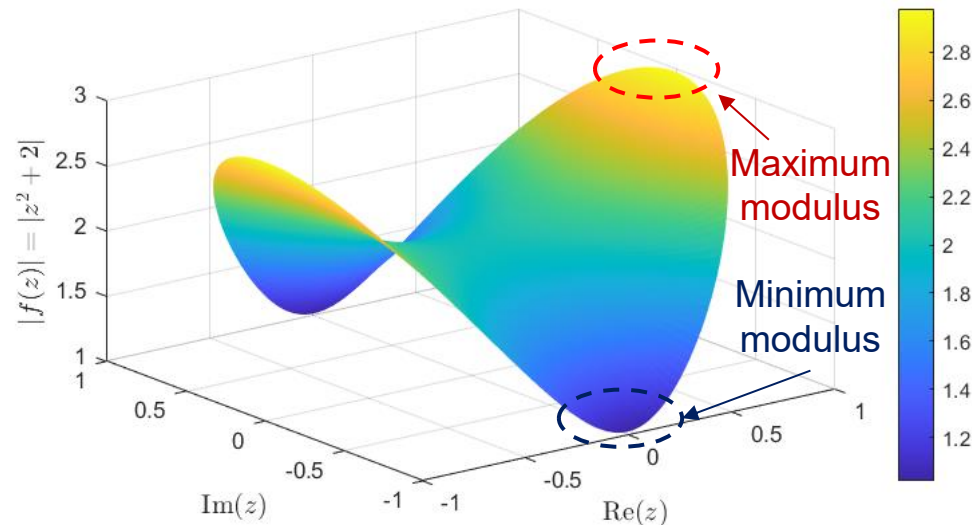
$$\text{LGC: } |(D_i(s))^{-1} + N_{ii}(s)| > \sum_{j \neq i} |N_{ij}(s)|, \forall s \in \Gamma$$



How to solve the LGC-derived complex-function modulus inequality in the prohibited domain Γ ?

Maximum Modulus Principle: If a complex function is **analytic** in a bounded closed domain, the **maximum** of the function's modulus is achieved solely on the **boundary**.

Minimum Modulus Principle: If a complex function is **analytic** and **non-zero** in a bounded closed domain, the **minimum** of the function's modulus is also achieved on the **boundary**.

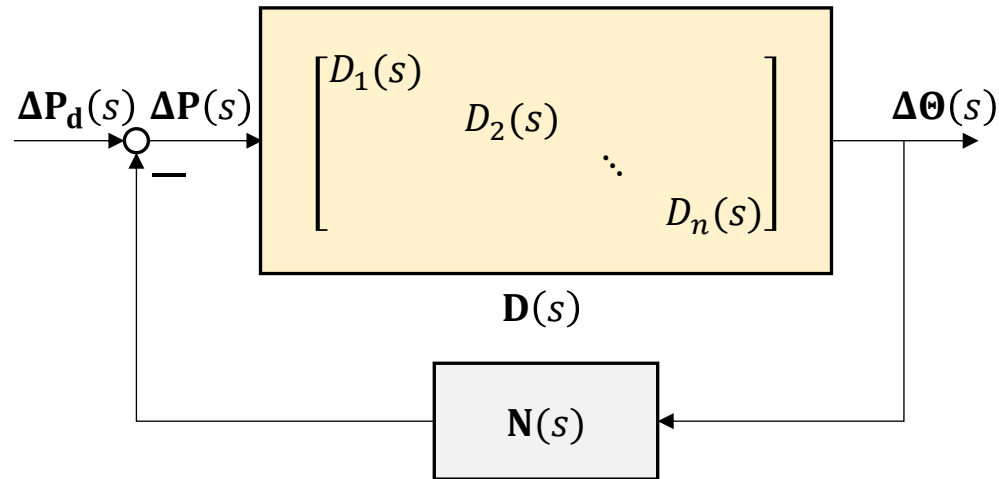


$$\max |f(s)| \Leftrightarrow \min \left| \frac{1}{f(s)} \right|$$

Remark: The infinite boundary can be regarded as a sufficiently large length R :

$$\text{RHP: } \{z \in \mathbb{C} | 0 \leq \text{Re}(z) \leq R, |\text{Im}(z)| \leq R, R \rightarrow \infty\}$$

Local Gain Boundary Condition (LGBC)



- LGBC:

$$\begin{cases} D_i^{-1}(s) + N_{ii}(s) \neq 0, \forall s \in \Gamma \\ |D_i^{-1}(s) + N_{ii}(s)| > \sum_{i \neq j} |N_{ij}(s)|, \forall s \in B \end{cases}$$

Sufficient
Condition

- LGC:

$$|(D_{ii}(s))^{-1} + N_{ii}(s)| > \sum_{i \neq j} |N_{ij}(s)|, \forall s \in \Gamma$$

Local Gain Boundary Condition (LGBC): If each diagonal element of $(D^{-1}(s) + N(s))$ is non-zero, then the verification of the LGC only needs to be performed **on the boundary of Γ** , e.g., $B = j\omega$.

LGBC vs. Small-Gain Condition

Example: Laplacian network N & $\Gamma = \text{RHP}$

Small-gain condition [2] and LGBC require $\mathbf{D}^{-1}(\mathbf{s})$ to fall outside their respective “circles”.

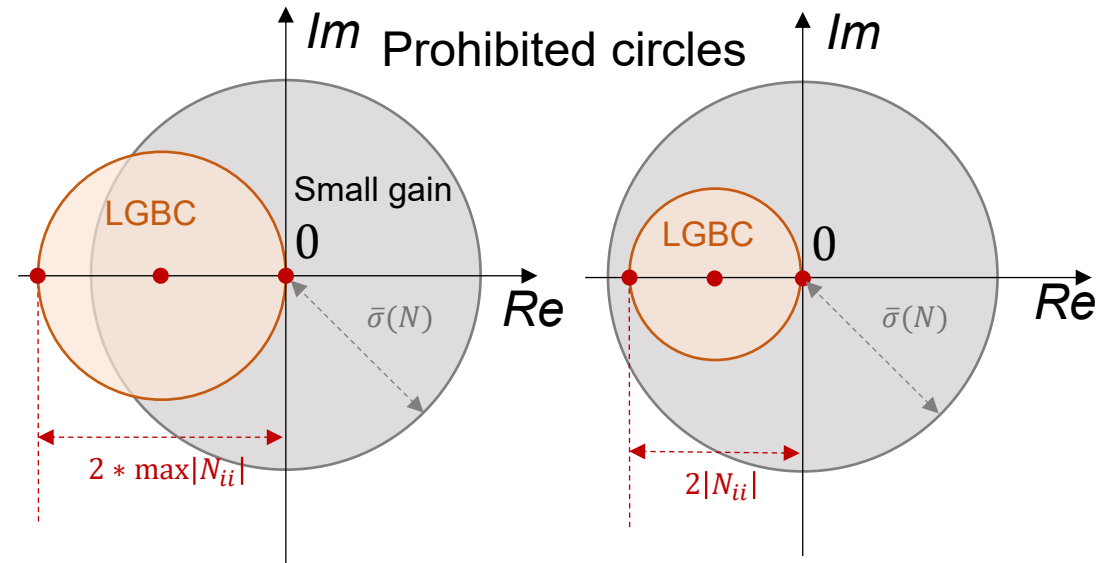
- Small-gain circle:
 $|D_i^{-1}(s)| > \max \text{ singular value } \bar{\sigma}(\mathbf{N}), \forall s = j\omega$

- LGBC circles:

$$|D_i^{-1}(s) + N_{ii}| > \sum_{j \neq i} |N_{ij}|, \forall s = j\omega$$

- Radius comparison:

$$\bar{\sigma}(\mathbf{N}) = \max(\lambda(\mathbf{N})) \geq \max(N_{ii})$$



Remark:

- The small-gain circle adopt a **"one-size-fits-all"** design
- LGBC circles features a **"locally tailored"** requirement, customized individually for each device
- The largest LGBC circle is **smaller than** the small-gain circle

LGBC vs. Small-Gain Condition / Passivity Condition

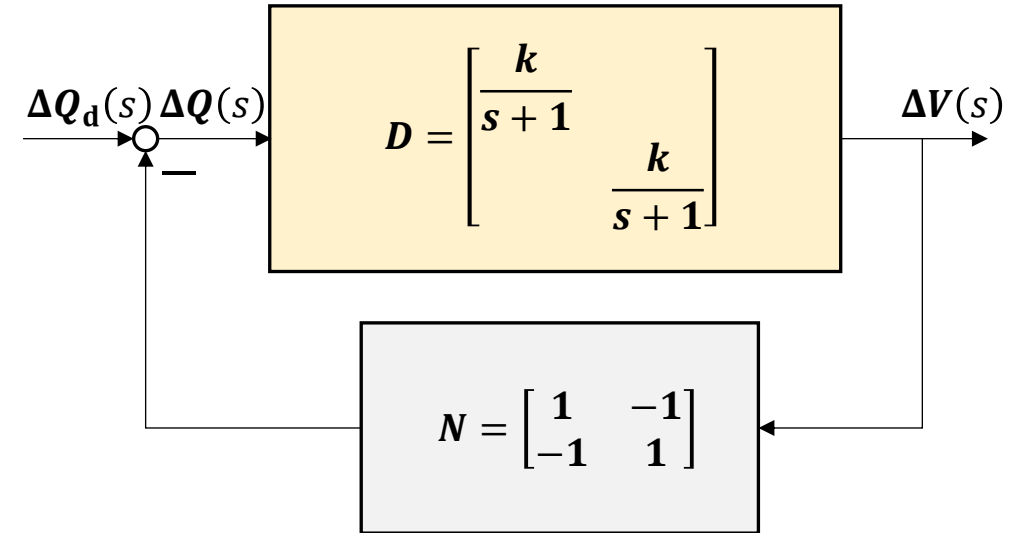
A toy example: Voltage stability with static Laplacian network N & $\Gamma = \text{RHP}$

- Closed-loop transfer function matrix:

$$G(s) = (I + DN)^{-1}D$$

$$= \frac{k}{(s + 2k + 1)(s + 1)} \begin{bmatrix} s + 1 + k & k \\ k & s + 1 + k \end{bmatrix}$$

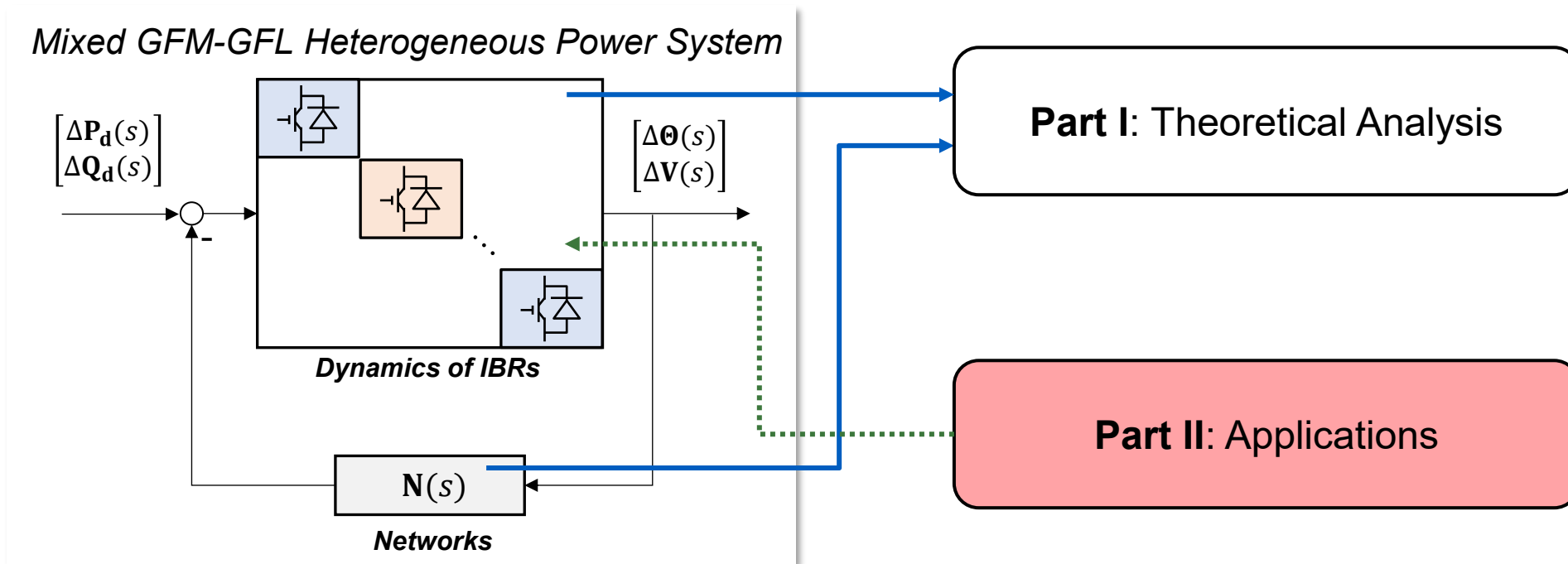
- Exact stability condition: $k > -\frac{1}{2}$
- Passivity-based stability condition: $k > 0$
- Small-gain-based stability condition: $2|k| < 1 \rightarrow -\frac{1}{2} < k < \frac{1}{2}$
- LGBC-based stability condition: $\begin{cases} k > -1 \\ \left| \frac{j\omega+1}{k} + 1 \right| > 1 \end{cases} \rightarrow k > -\frac{1}{2}$



LGBC:

$$\begin{cases} D_i^{-1}(s) + N_{ii}(s) \neq 0, \forall s \in \Gamma \\ |D_i^{-1}(s) + N_{ii}(s)| > \sum_{i \neq j} |N_{ij}(s)|, \forall s \in B \end{cases}$$

Outline



Definition of the Prohibited Domain

- Stability-defined prohibited domain (RHP):
Parameterize the LGBC on the imaginary $s = j\omega$ analytically

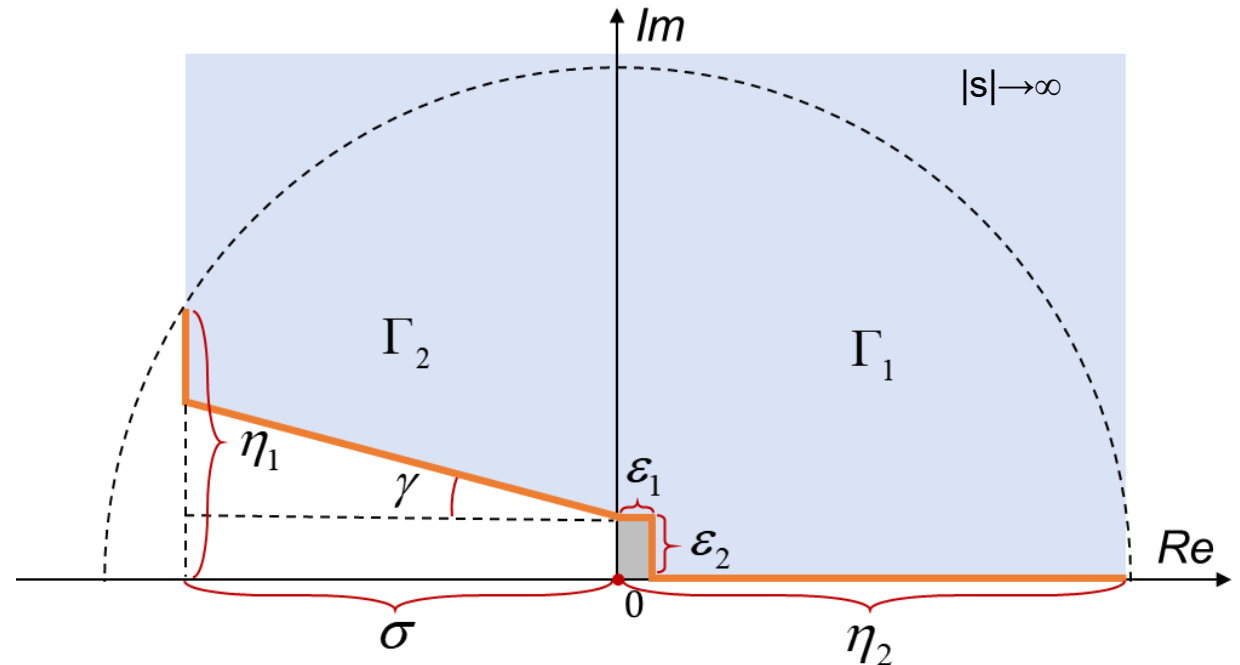
- Custom-defined prohibited domain:

$$\Gamma = \Gamma_1 \cup \Gamma_2$$

$$\begin{cases} \Gamma_1 = \{s | \operatorname{Re}(s) \geq 0, \operatorname{Im}(s) \geq 0, s \neq 0\} \\ \Gamma_2 = \left\{s | -\sigma \leq \operatorname{Re}(s) < 0, \frac{\operatorname{Im}(s)}{-\operatorname{Re}(s)} \geq \tan \gamma\right\} \end{cases}$$

Verify the LGBC on its finite boundary:

$$B^f = \begin{cases} \operatorname{Re}(s) = -\sigma, \sigma \tan \gamma + \varepsilon_2 \leq \operatorname{Im}(s) \leq \eta_1 \\ -\sigma \leq \operatorname{Re}(s) \leq 0, \operatorname{Im}(s) = \varepsilon_2 - \operatorname{Re}(s) \tan \gamma \\ 0 \leq \operatorname{Re}(s) \leq \varepsilon_1, \operatorname{Im}(s) = \varepsilon_2 \\ \operatorname{Re}(s) = \varepsilon_1, 0 \leq \operatorname{Im}(s) \leq \varepsilon_2 \\ \varepsilon_1 \leq \operatorname{Re}(s) \leq \eta_2, \operatorname{Im}(s) = 0 \end{cases}$$



Parameter Feasible Regions Determination

Algorithm 1 Parameter feasible region determination.

Input: Boundary $\mathcal{B}^f$, device matrix $\mathbf{D}(s)$ and network $\mathbf{N}(s)$.

Output: Parameter feasible regions $\mathcal{F} = \{\mathcal{F}_1, \dots, \mathcal{F}_n\}$.

```

1: Calculate parameter ranges  $\{\Xi_i\}_{i=1}^n$  that ensure
    $|D_i^{-1}(s, \Xi_i) + N_{ii}(s)| \neq 0, \text{Re}(s) > -\sigma$ .
2: Discretize boundary  $\mathcal{B}^f$  into point set  $\mathcal{S}$ .
3: for device  $i \leftarrow 1$  to  $n$  do
4:    $\mathcal{F}_i \leftarrow \emptyset$  and discretize  $\Xi_i$  into candidate set  $\Psi_i$ .
5:   for each parameter  $\psi \in \Psi_i$ . do
6:     if  $|D_i^{-1}(s, \psi) + N_{ii}(s)| > \sum_{j \neq i} |N_{ij}(s)|, \forall s \in \mathcal{S}$ .
       then
7:        $\mathcal{F}_i \leftarrow \mathcal{F}_i \cup \{\psi\}$ .
8:     end if
9:   end for
10: end for
11: return  $\mathcal{F}$ 

```

Parallel computation by IBRs



High efficiency regardless of scale

Converting Parameter Feasible Region to Constraint

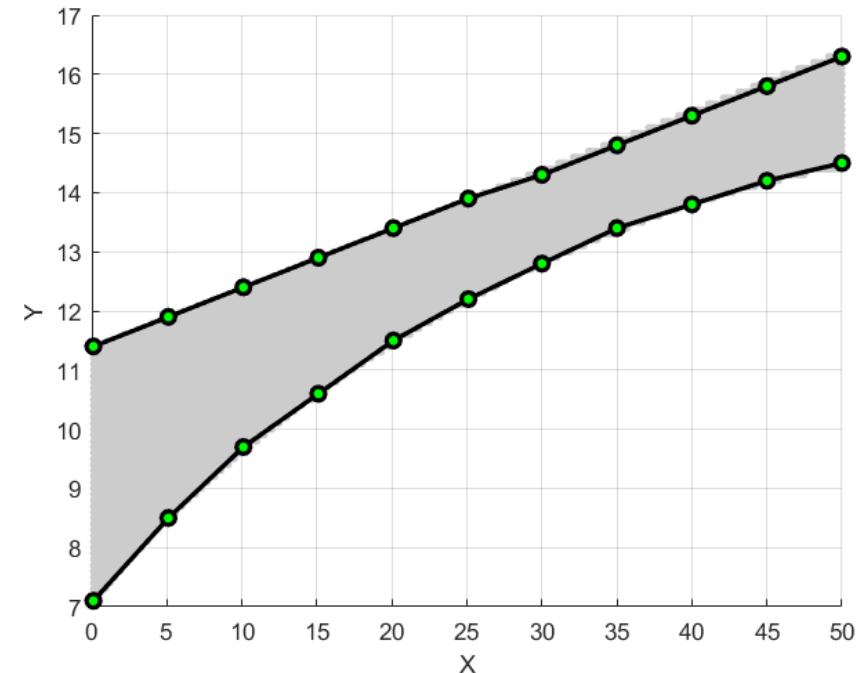
Main idea: Introduce non-negative variables to construct **mixed-integer linear constraints**, which can be directly applied to planning or scheduling problems.

- First, segment the parameter domain.
For each segment:
 - Abscissa vector: $X = (x_1 \ x_2 \ \cdots \ x_n)$
 - Upper bound vector of the ordinate: $Y^U = (y_1^u \ y_2^u \ \cdots \ y_n^u)$
 - Lower bound vector of the ordinate: $Y^L = (y_1^l \ y_2^l \ \cdots \ y_n^l)$
- Then construct the **linear constraint**:

$$\begin{cases} x = \sum_{i=1}^n \lambda_i x_i \\ \sum_{i=1}^n \lambda_i y_i^l \leq y \leq \sum_{i=1}^n \lambda_i y_i^u \\ \sum_{i=1}^n \lambda_i = 1, \text{SOS}(2) \\ \lambda \geq 0 \end{cases}$$



For parameter optimization application, etc.



SOS(2): At most two adjacent weights are non-zero

Two-IBR Case Study

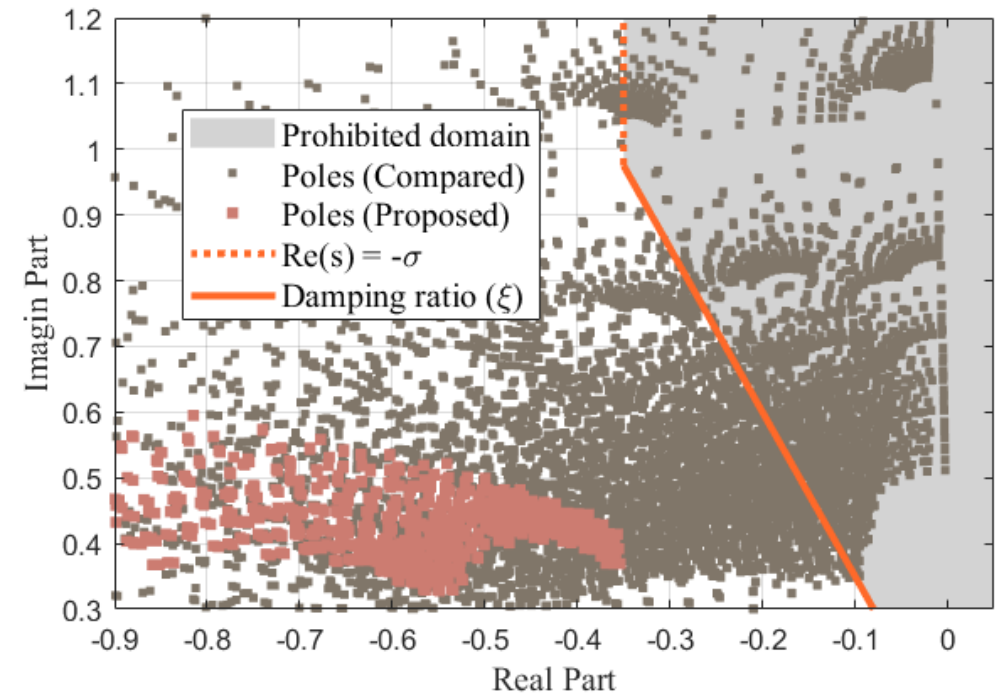
- Primarily focus on low-frequency oscillation (LFO)
- The dynamics of GFM and GFL are defined as:

$$D_i^g(s) = -\frac{1}{s(m_i s + d_i)}$$

$$\left(D_i^f(s)\right)^{-1} = -\frac{s^2 + V_0 K_i^P s + V_0 K_i^I}{s(H_i s + D_i)(V_0 K_i^P s + V_0 K_i^I)}$$

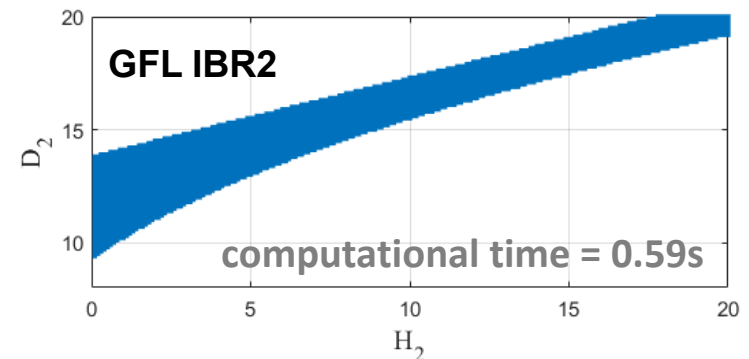
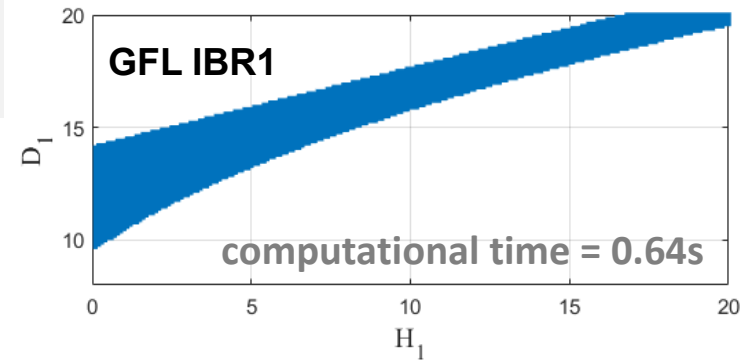
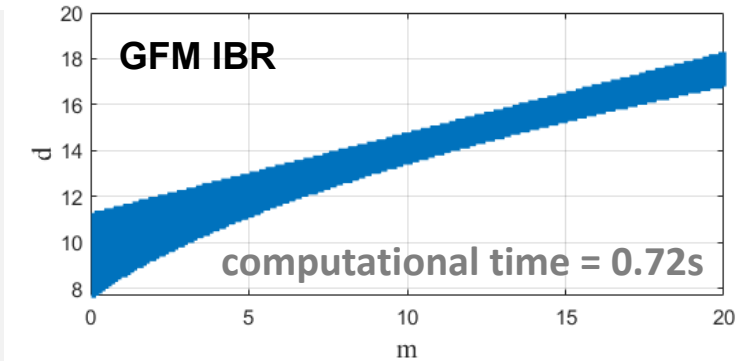
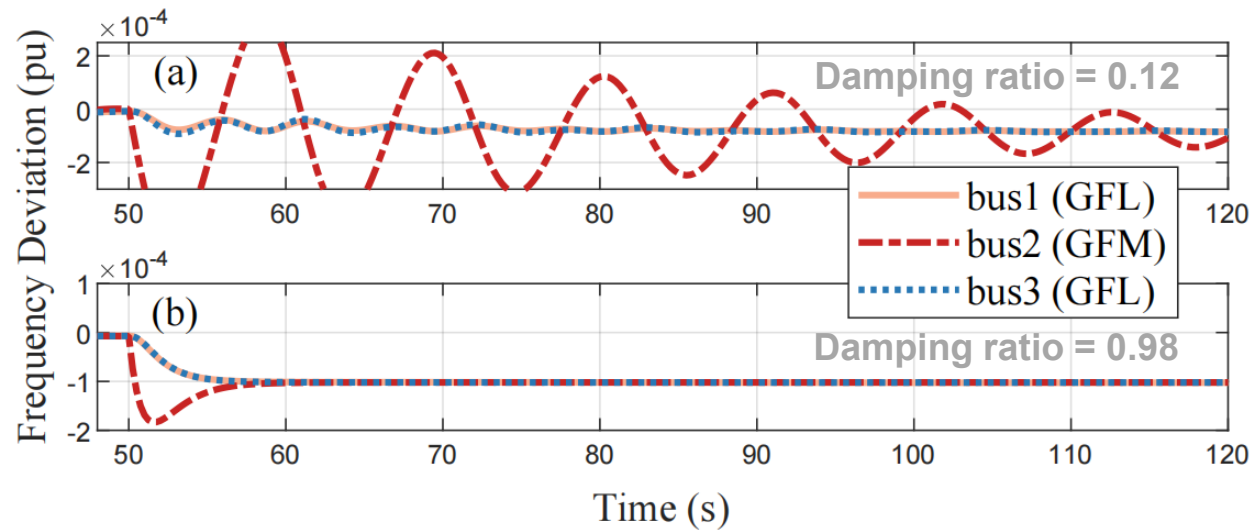
Parameter	Value	Parameter	Value	Parameter	Value
σ	0.35	η_1	10	H_1	0.1~20 (s ² /rad)
ξ	0.37	η_2	10	D_1	0.1~20 (s/rad)
K_1^P, K_1^I	4, 40	m	0.1~20 (s ² /rad)	H_2	0.1~20 (s ² /rad)
K_2^P, K_2^I	2, 20	d	0.1~20 (s/rad)	D_2	0.1~20 (s/rad)

- No poles lie within the predefined prohibited domain**
- Instability and weakly damped oscillations are avoided**



Three-IBR Case Study

- A three-IBR test system (one GFM IBR, two GFL IBR) is used to verify the proposed conditions
- The default parameters lead to weakly-damped poles lying in the prohibited domain (Fig (a))
- LGC-derived parameters ensure a required damping ratio (Fig (b))
- High efficiency for calculating parameter feasible regions of IBRs



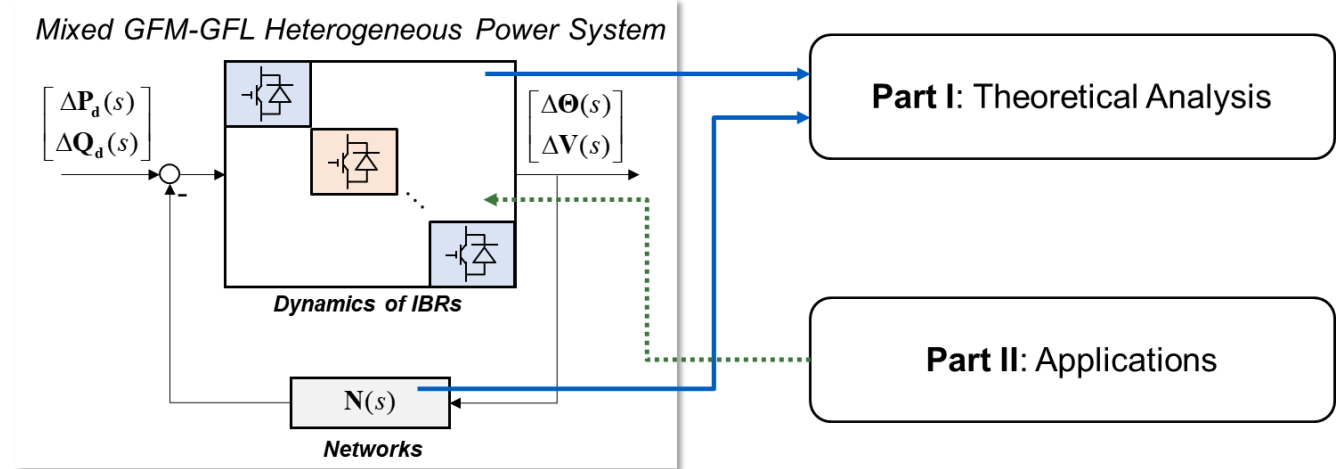
Conclusions and Future Work

- **Part I: Theoretical Analysis**

- ✓ Local gain conditions (LGC)
- ✓ Local gain boundary conditions (LGBC)

- **Part II: Applications**

- ✓ Definition of a prohibited domain
- ✓ Algorithm for determining the parameter feasible regions
- ✓ Quantifying as parametric constraints for optimization



Future Work

- Incorporate **voltage dynamics** in the modeling
- Incorporate phase information in the decentralized stability conditions, in addition to norm/magnitude information
- Compare with **other decentralized approaches (Small gains, small phases, SRGs, Davis–Wielandt shells)**
- Optimize frequency and voltage control with parametric stability conditions



Acknowledgement



Xiang Zhu
PhD Student
Tsinghua Univ.



Dr. Hongyang Qing
Postdoc
Tsinghua Univ.



Prof. Hua Geng
Full Professor
Tsinghua Univ.

X. Zhu, X. He, H. Qing, H. Geng. Decentralized Analysis Approach for Oscillation Damping in Grid-Forming and Grid-Following Heterogeneous Power Systems. arXiv Preprint, 2026, <https://doi.org/10.48550/arXiv.2601.21562>.



清华大学
Tsinghua University

Thanks Q & A

Xiuqiang He

xhe@tsinghua.edu.cn